

## # Abir Hasan Khan - Professional Profile & Engineering Resume

### ## Metadata & Contact

- **Full Name**: Abir Hasan Khan
- **Primary Role**: Full Stack Developer / Software Engineer
- **Target Roles**: Full Stack Developer, Frontend Engineer, Backend Engineer
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- **Location**: Dhaka, Bangladesh
- **Profiles & Links**:
  - LinkedIn: <https://www.linkedin.com/in/abir-hasan-khan/>
  - GitHub: <https://github.com/shaonexplorer>
  - Portfolio: <https://abir-khan-portfolio.netlify.app/>

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### ## Professional Summary

Abir Hasan Khan is a Professional Full Stack Developer with hands-on expertise in building efficient, type-safe, and high-performance web applications. Skilled across modern web technologies including Next.js, Node.js, TypeScript, PostgreSQL, and Express.js. Experienced in optimizing database queries, building event-driven architectures, integrating state management solutions, and deploying AI-driven utilities to maximize operational efficiency.

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### ## Core Technical Skills & Competencies

#### ### Programming Languages

- TypeScript
- JavaScript (ES6+)
- SQL
- HTML5
- CSS3

#### ### Frontend Development

- **Frameworks & Libraries**: React.js, Next.js (App Router, Server Components)
- **State Management & Data Fetching**: Redux Toolkit, TanStack Query (React Query v5)
- **UI & Styling**: Tailwind CSS, shadcn/ui, TanStack Table (v8)
- **Form Handling & Validation**: React Hook Form, Zod

#### ### Backend & Database Architecture

- **Runtime & Frameworks**: Node.js, Express.js (Express 5)

- **Databases & ORMs**: PostgreSQL, Prisma ORM, MongoDB, Mongoose ORM
- **API Design & Protocols**: RESTful APIs, WebSockets (Socket.io)

### ### AI, DevOps & Tooling

- **AI Integration**: AI SDK, Retrieval-Augmented Generation (RAG), OpenAI API, Model Context Protocol (MCP)
- **Version Control & CI/CD**: Git, GitHub, GitHub Actions
- **Deployment & Cloud Platforms**: Vercel, Render

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## ## Professional Work Experience

### ### Jr MERN Developer

**Company**: Softvence Agency

**Duration**: January 2026 - May 2026

**Key Accomplishments & Responsibilities**:

- **Performance Optimization**: Developed full-stack capabilities leveraging Next.js App Router and Server Components, cutting client bundle size by ~30% and significantly enhancing Core Web Vitals.
- **UI Component Architecture**: Translated complex Figma designs into modular, re-usable, and fully type-safe React/shadcn UI components across multiple fast-paced development sprints.
- **API Development**: Designed, implemented, and supported robust RESTful API endpoints utilizing Node.js, PostgreSQL, and Prisma ORM.

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## ## Technical Projects & Software Architecture

### ### 1. Full-Stack Pharmacy & Inventory Management System

- **Project Overview**: A high-reliability enterprise application handling real-time point-of-sale transactions and inventory management.
- **Transactional POS Terminal**: Architected a multi-step checkout workflow in Express.js using Prisma database transactions (`\$transaction`) to guarantee atomic stock decrements, ledger creation, and order records in a single round-trip database call.
- **Server-Side Data Grid Optimization**: Integrated TanStack Table v8 with TanStack Query (v5) leveraging the `keepPreviousData` caching strategy to enable seamless pagination, dynamic filtering, and zero UI layout shifts during transitions.
- **Spec-Driven Monorepo Architecture**: Structured a modular Turborepo monorepo with centralized Zod validation schemas and shared TypeScript type packages across Next.js 16 (App Router) and Express.js 5 for complete end-to-end type safety.

- **Tech Stack**: React.js, Next.js 16, TypeScript, Express.js 5, PostgreSQL, Prisma ORM, TanStack Table v8, TanStack Query v5, Zod, JWT Authentication.

### 2. High-Performance Multi-Tenant Board System

- **Project Overview**: A scalable collaborative board management application with interactive drag-and-drop workflow engines.
- **Fractional Ordering Engine**: Implemented a logarithmic float positioning algorithm for column and task ordering, enabling  $O(1)$  time complexity for drag-and-drop reordering without requiring costly database re-indexing locks.
- **Responsive Multi-Tier Layout**: Built a tier-aware responsive user interface using Next.js 16 and `@dnd-kit`, rendering touch-optimized tab flows on mobile devices and pointer drag-and-drop mechanics on desktop screens.
- **Automated Security Audit**: Developed an Express.js 5 and Zod validation framework coupled with a custom route auditor script to ensure 100% input validation coverage across all private API endpoints.
- **Tech Stack**: Next.js 16, TypeScript, Express.js 5, PostgreSQL, TanStack Query, React Hook Form, Zod, `@dnd-kit`, JWT Authentication.

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### Education, Training & Certifications

- **Next Level Web Development Bootcamp** – Programming Hero
- **AI Engineering Path** – Scrimba
- **Claude Code for Professional Developers** – Code with Mosh